SoccerGMR Standard Test - GMR / Multi methods

Test value with (delta vs GMR baseline) below | G-mIoU: 0-1 | Other values / deltas: % / percentage points

		Threshold-independent (%)					Threshold = 0.4				Threshold = 0.5				Threshold = 0.6			
Backbone	**Method**	**AUROC**	**mAP**	**mR@1**	**mR@5**	**mR+@5**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**	**N-acc**	**Rej-F1**	**G-mIoU@1**	**G-mIoU@3**
Moment-DETR	**GMR (baseline)**	**68.80**	**5.91**	**3.68**	**7.98**	**1.16**	**0.00**	**0.00**	**0.0577**	**0.0220**	**100.00**	**75.00**	**0.5239**	**0.4912**	**100.00**	**75.00**	**0.5239**	**0.4912**
	GMR + GEQ Frozen	68.18 **(-0.62)**	5.91 (0.00)	3.68 (0.00)	7.98 (0.00)	1.16 (0.00)	100.00 **(+100.00)**	75.00 **(+75.00)**	0.5239 **(+0.4662)**	0.4912 **(+0.4692)**	100.00 (0.00)	75.00 (0.00)	0.5239 (0.0000)	0.4912 (0.0000)	100.00 (0.00)	75.00 (0.00)	0.5239 (0.0000)	0.4912 (0.0000)
	GMR + GEQ Full	68.96 **(+0.16)**	6.09 **(+0.18)**	3.77 **(+0.09)**	8.19 **(+0.21)**	0.66 **(-0.50)**	99.59 **(+99.59)**	74.81 **(+74.81)**	0.5243 **(+0.4666)**	0.4901 **(+0.4681)**	100.00 (0.00)	75.00 (0.00)	0.5263 **(+0.0024)**	0.4920 **(+0.0008)**	100.00 (0.00)	75.00 (0.00)	0.5263 **(+0.0024)**	0.4920 **(+0.0008)**
	GMR + Multi	70.14 **(+1.34)**	7.27 **(+1.36)**	4.71 **(+1.03)**	10.04 **(+2.06)**	0.28 **(-0.88)**	83.54 **(+83.54)**	69.60 **(+69.60)**	0.4421 **(+0.3844)**	0.4136 **(+0.3916)**	100.00 (0.00)	75.00 (0.00)	0.5188 **(-0.0051)**	0.4895 **(-0.0017)**	100.00 (0.00)	75.00 (0.00)	0.5188 **(-0.0051)**	0.4895 **(-0.0017)**
	GMR + Multi + GEQ Frozen	69.25 **(+0.45)**	7.27 **(+1.36)**	4.71 **(+1.03)**	10.04 **(+2.06)**	0.28 **(-0.88)**	90.65 **(+90.65)**	72.11 **(+72.11)**	0.4749 **(+0.4172)**	0.4456 **(+0.4236)**	100.00 (0.00)	75.00 (0.00)	0.5188 **(-0.0051)**	0.4895 **(-0.0017)**	100.00 (0.00)	75.00 (0.00)	0.5188 **(-0.0051)**	0.4895 **(-0.0017)**
	GMR + Multi + GEQ Full	69.83 **(+1.03)**	7.72 **(+1.81)**	4.06 **(+0.38)**	12.48 **(+4.50)**	0.83 **(-0.33)**	81.91 **(+81.91)**	68.89 **(+68.89)**	0.4395 **(+0.3818)**	0.4070 **(+0.3850)**	100.00 (0.00)	75.00 (0.00)	0.5237 **(-0.0002)**	0.4912 (0.0000)	100.00 (0.00)	75.00 (0.00)	0.5237 **(-0.0002)**	0.4912 (0.0000)
EaTR	**GMR (baseline)**	**76.56**	**16.20**	**10.64**	**21.38**	**9.59**	**48.58**	**57.11**	**0.3332**	**0.2834**	**66.67**	**65.60**	**0.4085**	**0.3600**	**87.20**	**73.46**	**0.4956**	**0.4486**
	GMR + GEQ Frozen	75.17 **(-1.39)**	16.20 (0.00)	10.64 (0.00)	21.38 (0.00)	9.59 (0.00)	63.82 **(+15.24)**	64.02 **(+6.91)**	0.3911 **(+0.0579)**	0.3452 **(+0.0618)**	87.60 **(+20.93)**	72.93 **(+7.33)**	0.4933 **(+0.0848)**	0.4489 **(+0.0889)**	96.75 **(+9.55)**	74.90 **(+1.44)**	0.5315 **(+0.0359)**	0.4858 **(+0.0372)**
	GMR + GEQ Full	75.26 **(-1.30)**	17.15 **(+0.95)**	11.47 **(+0.83)**	22.84 **(+1.46)**	10.79 **(+1.20)**	64.23 **(+15.65)**	64.29 **(+7.18)**	0.3988 **(+0.0656)**	0.3482 **(+0.0648)**	90.65 **(+23.98)**	74.77 **(+9.17)**	0.5111 **(+0.1026)**	0.4635 **(+0.1035)**	98.17 **(+10.98)**	75.23 **(+1.77)**	0.5402 **(+0.0446)**	0.4921 **(+0.0435)**
	GMR + Multi	77.11 **(+0.55)**	17.52 **(+1.32)**	11.64 **(+1.00)**	23.70 **(+2.32)**	11.77 **(+2.18)**	44.92 **(-3.66)**	55.18 **(-1.93)**	0.3305 **(-0.0027)**	0.2702 **(-0.0132)**	61.59 **(-5.08)**	63.79 **(-1.81)**	0.3988 **(-0.0097)**	0.3424 **(-0.0176)**	75.41 **(-11.79)**	69.15 **(-4.31)**	0.4575 **(-0.0381)**	0.4012 **(-0.0474)**
	GMR + Multi + GEQ Frozen	76.69 **(+0.13)**	17.52 **(+1.32)**	11.64 **(+1.00)**	23.70 **(+2.32)**	11.77 **(+2.18)**	69.51 **(+20.93)**	68.13 **(+11.02)**	0.4277 **(+0.0945)**	0.3738 **(+0.0904)**	81.50 **(+14.84)**	71.04 **(+5.44)**	0.4738 **(+0.0653)**	0.4243 **(+0.0643)**	89.63 **(+2.44)**	73.01 **(-0.45)**	0.5058 **(+0.0102)**	0.4571 **(+0.0085)**
	GMR + Multi + GEQ Full	77.27 **(+0.71)**	19.88 **(+3.68)**	14.05 **(+3.41)**	25.31 **(+3.93)**	12.24 **(+2.65)**	68.90 **(+20.33)**	67.60 **(+10.49)**	0.4318 **(+0.0986)**	0.3741 **(+0.0907)**	81.30 **(+14.63)**	71.05 **(+5.45)**	0.4796 **(+0.0711)**	0.4260 **(+0.0660)**	90.65 **(+3.46)**	73.48 **(+0.02)**	0.5165 **(+0.0209)**	0.4632 **(+0.0146)**
QD-DETR	**GMR (baseline)**	**77.66**	**15.08**	**10.04**	**20.11**	**9.62**	**53.46**	**59.84**	**0.3533**	**0.3035**	**90.04**	**73.77**	**0.5058**	**0.4611**	**97.56**	**75.29**	**0.5355**	**0.4901**
	GMR + GEQ Frozen	77.42 **(-0.24)**	15.08 (0.00)	10.04 (0.00)	20.11 (0.00)	9.62 (0.00)	77.24 **(+23.78)**	70.11 **(+10.27)**	0.4559 **(+0.1026)**	0.4070 **(+0.1035)**	92.48 **(+2.44)**	74.47 **(+0.70)**	0.5161 **(+0.0103)**	0.4711 **(+0.0100)**	96.95 **(-0.61)**	74.88 **(-0.41)**	0.5322 **(-0.0033)**	0.4868 **(-0.0033)**
	GMR + GEQ Full	77.66 (0.00)	16.54 **(+1.46)**	11.48 **(+1.44)**	22.09 **(+1.98)**	10.14 **(+0.52)**	80.49 **(+27.03)**	71.35 **(+11.51)**	0.4727 **(+0.1194)**	0.4232 **(+0.1197)**	93.90 **(+3.86)**	74.52 **(+0.75)**	0.5241 **(+0.0183)**	0.4769 **(+0.0158)**	98.17 **(+0.61)**	75.23 **(-0.06)**	0.5419 **(+0.0064)**	0.4942 **(+0.0041)**
	GMR + Multi	75.56 **(-2.10)**	14.67 **(-0.41)**	9.95 **(-0.09)**	19.50 **(-0.61)**	5.60 **(-4.02)**	53.86 **(+0.41)**	58.37 **(-1.47)**	0.3485 **(-0.0048)**	0.2985 **(-0.0050)**	91.67 **(+1.63)**	74.92 **(+1.15)**	0.5157 **(+0.0099)**	0.4690 **(+0.0079)**	96.54 **(-1.02)**	75.28 **(-0.01)**	0.5308 **(-0.0047)**	0.4846 **(-0.0055)**
	GMR + Multi + GEQ Frozen	74.92 **(-2.74)**	14.67 **(-0.41)**	9.95 **(-0.09)**	19.50 **(-0.61)**	5.60 **(-4.02)**	74.59 **(+21.14)**	68.41 **(+8.57)**	0.4380 **(+0.0847)**	0.3917 **(+0.0882)**	94.51 **(+4.47)**	74.82 **(+1.05)**	0.5224 **(+0.0166)**	0.4772 **(+0.0161)**	97.97 **(+0.41)**	74.67 **(-0.62)**	0.5320 **(-0.0035)**	0.4886 **(-0.0015)**
	GMR + Multi + GEQ Full	74.79 **(-2.87)**	13.71 **(-1.37)**	9.21 **(-0.83)**	18.42 **(-1.69)**	6.44 **(-3.18)**	94.31 **(+40.85)**	74.72 **(+14.88)**	0.5194 **(+0.1661)**	0.4756 **(+0.1721)**	98.98 **(+8.94)**	75.15 **(+1.38)**	0.5368 **(+0.0310)**	0.4932 **(+0.0321)**	99.59 **(+2.03)**	74.87 **(-0.42)**	0.5387 **(+0.0032)**	0.4951 **(+0.0050)**
CG-DETR	**GMR (baseline)**	**75.97**	**16.09**	**10.76**	**22.12**	**9.48**	**96.75**	**75.68**	**0.5382**	**0.4890**	**99.19**	**75.84**	**0.5465**	**0.4975**	**99.59**	**75.91**	**0.5484**	**0.4994**
	GMR + GEQ Frozen	76.39 **(+0.42)**	16.09 (0.00)	10.76 (0.00)	22.12 (0.00)	9.48 (0.00)	75.00 **(-21.75)**	69.04 **(-6.64)**	0.4457 **(-0.0925)**	0.3969 **(-0.0921)**	92.48 **(-6.71)**	74.23 **(-1.61)**	0.5189 **(-0.0276)**	0.4698 **(-0.0277)**	99.19 **(-0.41)**	75.95 **(+0.04)**	0.5463 **(-0.0021)**	0.4976 **(-0.0018)**
	GMR + GEQ Full	74.83 **(-1.14)**	16.58 **(+0.49)**	11.12 **(+0.36)**	22.60 **(+0.48)**	10.97 **(+1.49)**	75.00 **(-21.75)**	68.59 **(-7.09)**	0.4439 **(-0.0943)**	0.3933 **(-0.0957)**	97.36 **(-1.83)**	75.55 **(-0.29)**	0.5381 **(-0.0084)**	0.4889 **(-0.0086)**	99.80 **(+0.20)**	75.42 **(-0.49)**	0.5446 **(-0.0038)**	0.4978 **(-0.0016)**
	GMR + Multi	77.40 **(+1.43)**	16.59 **(+0.50)**	11.21 **(+0.45)**	21.77 **(-0.35)**	8.14 **(-1.34)**	46.95 **(-49.80)**	56.20 **(-19.48)**	0.3369 **(-0.2013)**	0.2783 **(-0.2107)**	75.41 **(-23.78)**	69.61 **(-6.23)**	0.4542 **(-0.0923)**	0.4005 **(-0.0970)**	90.45 **(-9.15)**	74.23 **(-1.68)**	0.5167 **(-0.0317)**	0.4641 **(-0.0353)**
	GMR + Multi + GEQ Frozen	75.95 **(-0.02)**	16.59 **(+0.50)**	11.21 **(+0.45)**	21.77 **(-0.35)**	8.14 **(-1.34)**	70.33 **(-26.42)**	67.32 **(-8.36)**	0.4310 **(-0.1072)**	0.3758 **(-0.1132)**	85.16 **(-14.02)**	72.68 **(-3.16)**	0.4938 **(-0.0527)**	0.4417 **(-0.0558)**	92.68 **(-6.91)**	73.91 **(-2.00)**	0.5198 **(-0.0286)**	0.4701 **(-0.0293)**
	GMR + Multi + GEQ Full	75.77 **(-0.20)**	18.59 **(+2.50)**	12.31 **(+1.55)**	24.98 **(+2.86)**	12.13 **(+2.65)**	59.15 **(-37.60)**	62.72 **(-12.96)**	0.3884 **(-0.1498)**	0.3304 **(-0.1586)**	81.30 **(-17.89)**	71.43 **(-4.41)**	0.4784 **(-0.0681)**	0.4246 **(-0.0729)**	93.90 **(-5.69)**	74.04 **(-1.87)**	0.5244 **(-0.0240)**	0.4750 **(-0.0244)**

Delta = method - GMR baseline in the SAME dataset / Test split / protocol / backbone.

Green: gain Red: decrease Gray: unchanged. Deltas use source precision, then round for display. N/A: not applicable.

GEQ checkpoints selected by Val G-mIoU@1 mean at 0.4 / 0.5 / 0.6. Native parent training budgets differ.

N-acc = TN / negative queries; Rej-F1 treats rejection as the target class. Threshold = existence-score threshold.

Source branch: gmr-mr-detr-multiroute-20261030 | Base: 22a882d45eb9 + verified supplemental Test exports